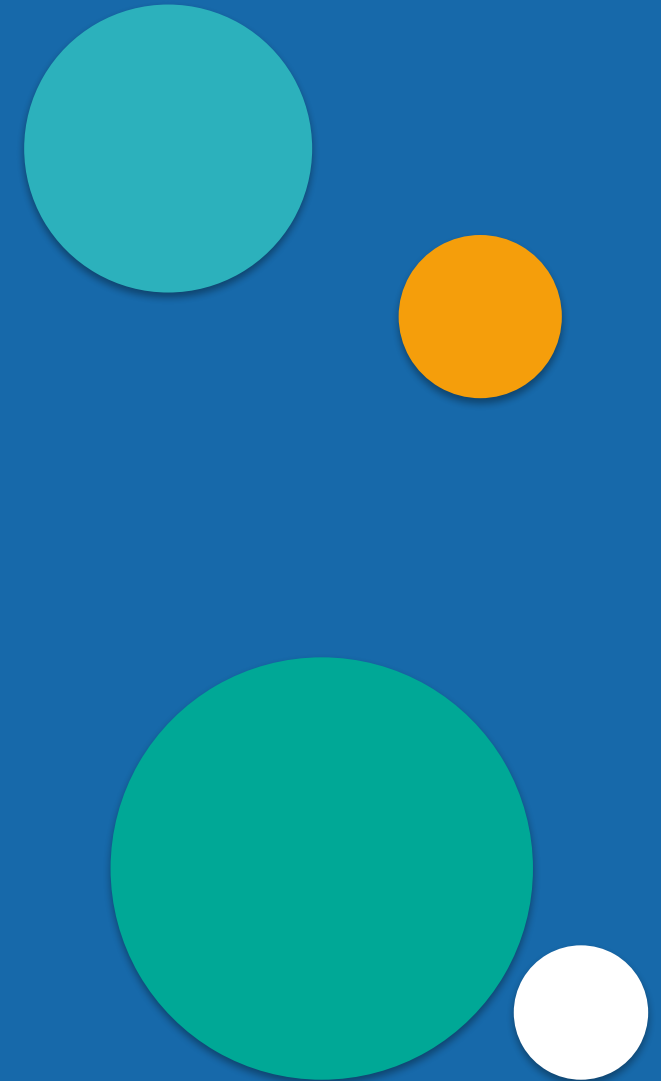


Retain the Right Customers Before They Leave

An AI-powered churn prevention proposal built from 100,000 customer records

Recommendation: rank customers weekly and focus retention actions on the highest-risk 20%.



Executive recommendation

Move from broad retention campaigns to risk-ranked, action-specific outreach

0.702

Held-out ROC-AUC

20,000-customer test set

30.0%

Churners captured

by targeting the top 20%

1.50×

Risk lift

74.4% vs 49.6% baseline

\$1.65M

Illustrative net value

per 100k customers/year

What Company A should do

- Score active customers weekly using usage, billing, handset and service-quality signals.
- Prioritize the highest-risk 20% for targeted offers, service recovery or handset upgrades.
- Measure incremental retention through a randomized control group before full rollout.

Why now

- The data already contains predictive behavior before churn.
- The solution is deployable as a probability-ranked workflow, not a one-time model.
- The business case remains positive under conservative intervention assumptions.

Retention is a telecom growth problem

Saturated markets increase the value of protecting the installed customer base

The strategic tension

Telecom operators must defend revenue while customers face low switching friction, aggressive offers and rising service expectations.

Industry signal

22% average annual churn across providers in Deloitte's 2026 outlook.

Company A's response should be selective: identify preventable churn early and spend only where intervention is economically justified.

Revenue risk

Customers leave with recurring revenue and future lifetime value.

Service signals

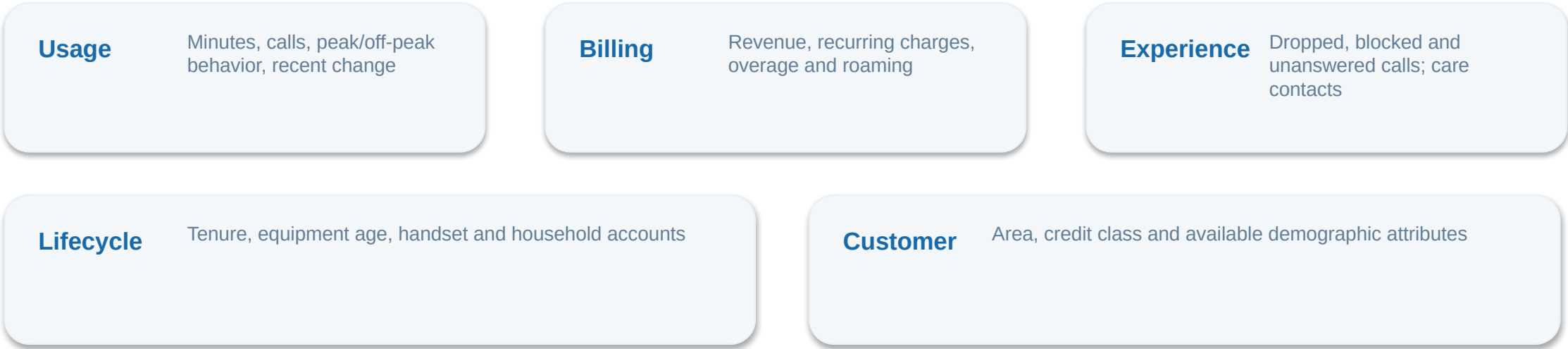
Usage decline, failures and support activity can reveal friction.

Action window

The target records churn 31–60 days after observation.

Business question: who should receive which retention action, and when?

Company A already has the data required for a retention PoC

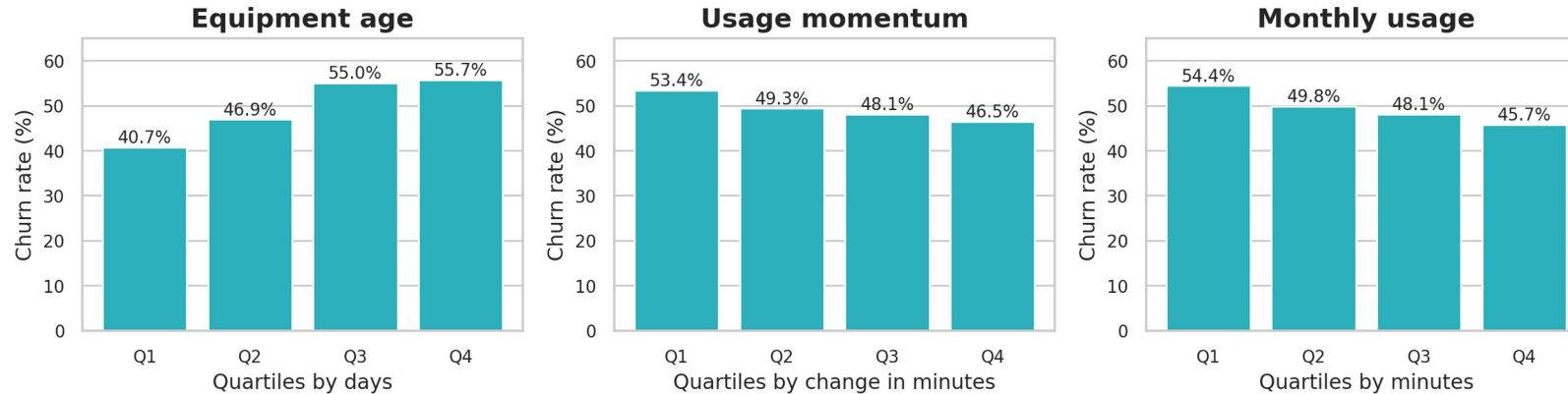


Important caveat: the supplied target is almost 50/50 churn, unlike many live telecom portfolios. Production calibration must use Company A's current base rate.

EDA reveals actionable warning signals

Single variables are insufficient, but several behaviors show consistent directional risk

Churn changes materially across actionable customer signals



What the patterns say

- Older equipment is associated with higher churn.
- Declining usage is an early warning signal.
- Low current usage indicates weaker engagement.
- No single variable separates churners; multivariate modeling is required.

Actionable hypothesis

Combine lifecycle, engagement and service signals to trigger targeted actions.

Problem statement and success definition

Business problem

Company A cannot economically treat every customer, but it lacks a repeatable method to identify who is most likely to churn and which operational lever should be used.

ML objective

Rank customers by probability of churn 31–60 days ahead.

Business objective

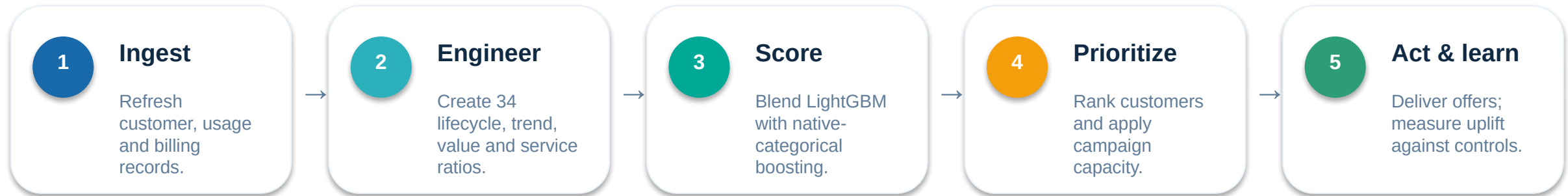
Concentrate intervention spend where avoidable revenue risk is highest.

Success metrics

ROC-AUC and PR-AUC for ranking; capture, lift and incremental retention for business value.

Decision rule for the PoC: use risk ranking to define campaign capacity; do not treat the model threshold as an automatic cancellation or denial decision.

Solution: a weekly risk-ranking and action engine



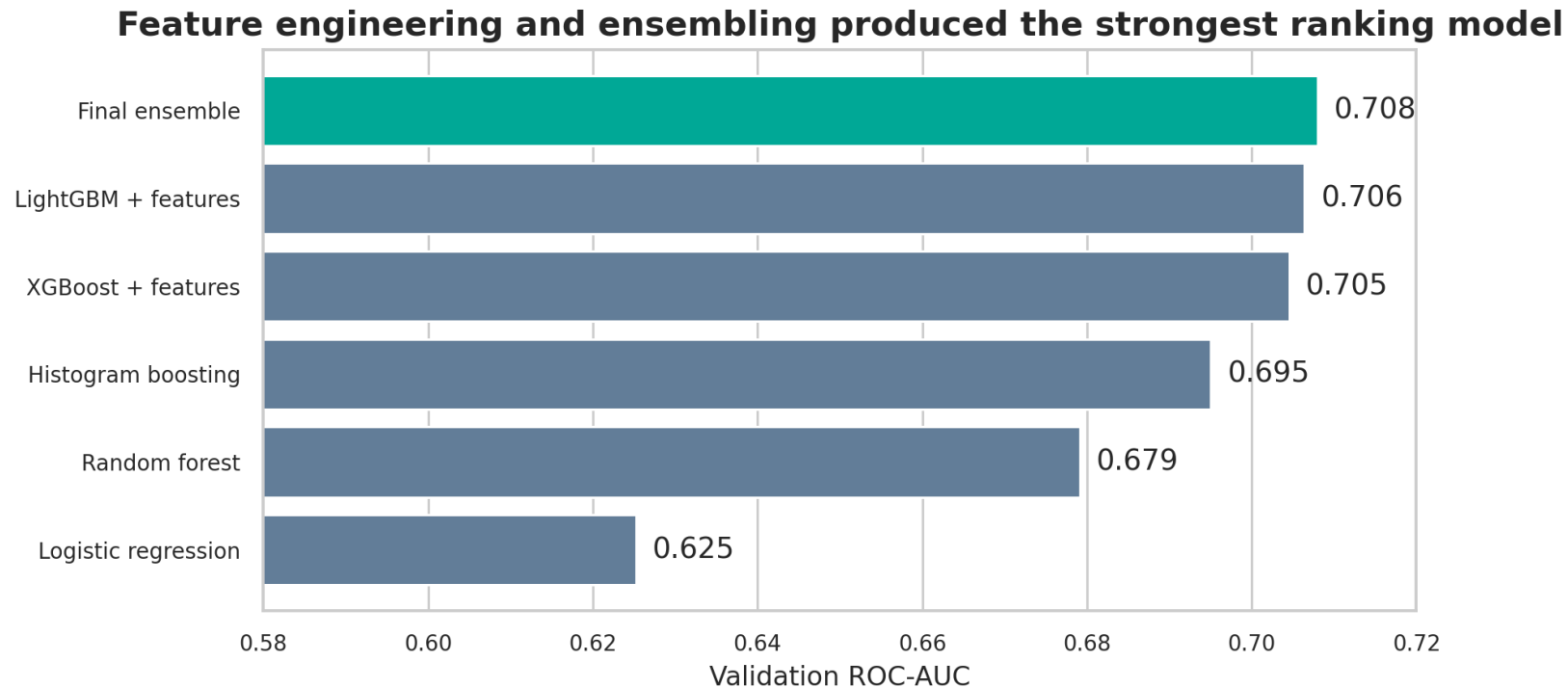
Why this design is practical

- Works on existing tabular data; no new platform is required for the PoC.
- Produces probabilities that support multiple campaign capacities and cost structures.

Controls built into the workflow

- Human-approved action catalogue; no adverse automated decisions.
- Monitoring for drift, calibration, fairness and incremental campaign lift.

Model development moved beyond the tutorial baseline



Final model

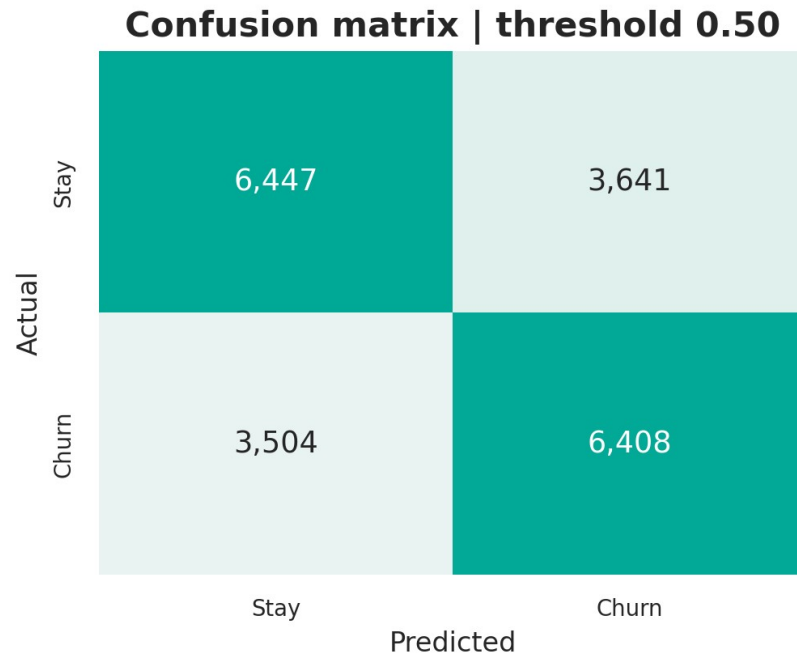
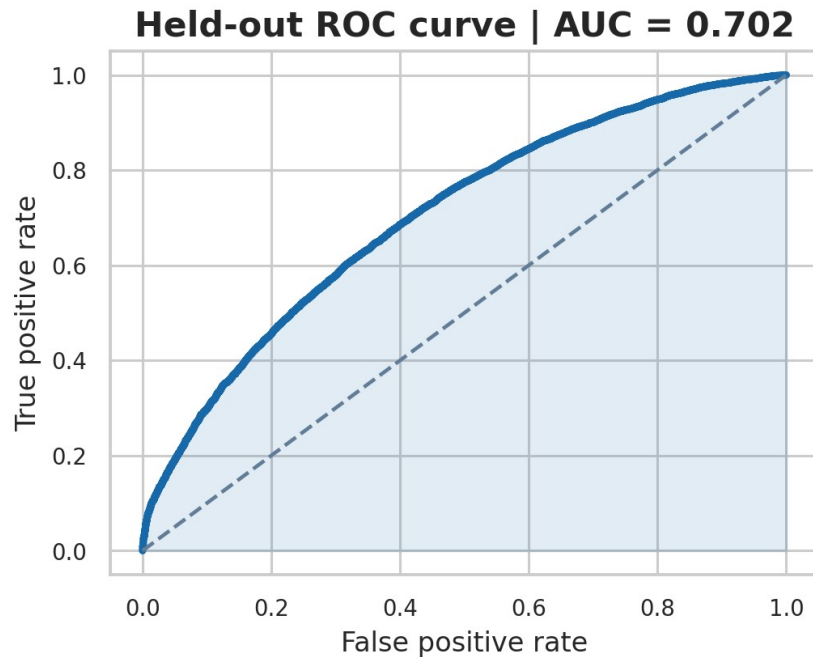
65% Engineered
LightGBM

35% Categorical
ensemble

Selection protocol

- 60% train
- 20% validation
- 20% untouched test
- Selection by validation ROC-AUC

The final model generalizes to unseen customers



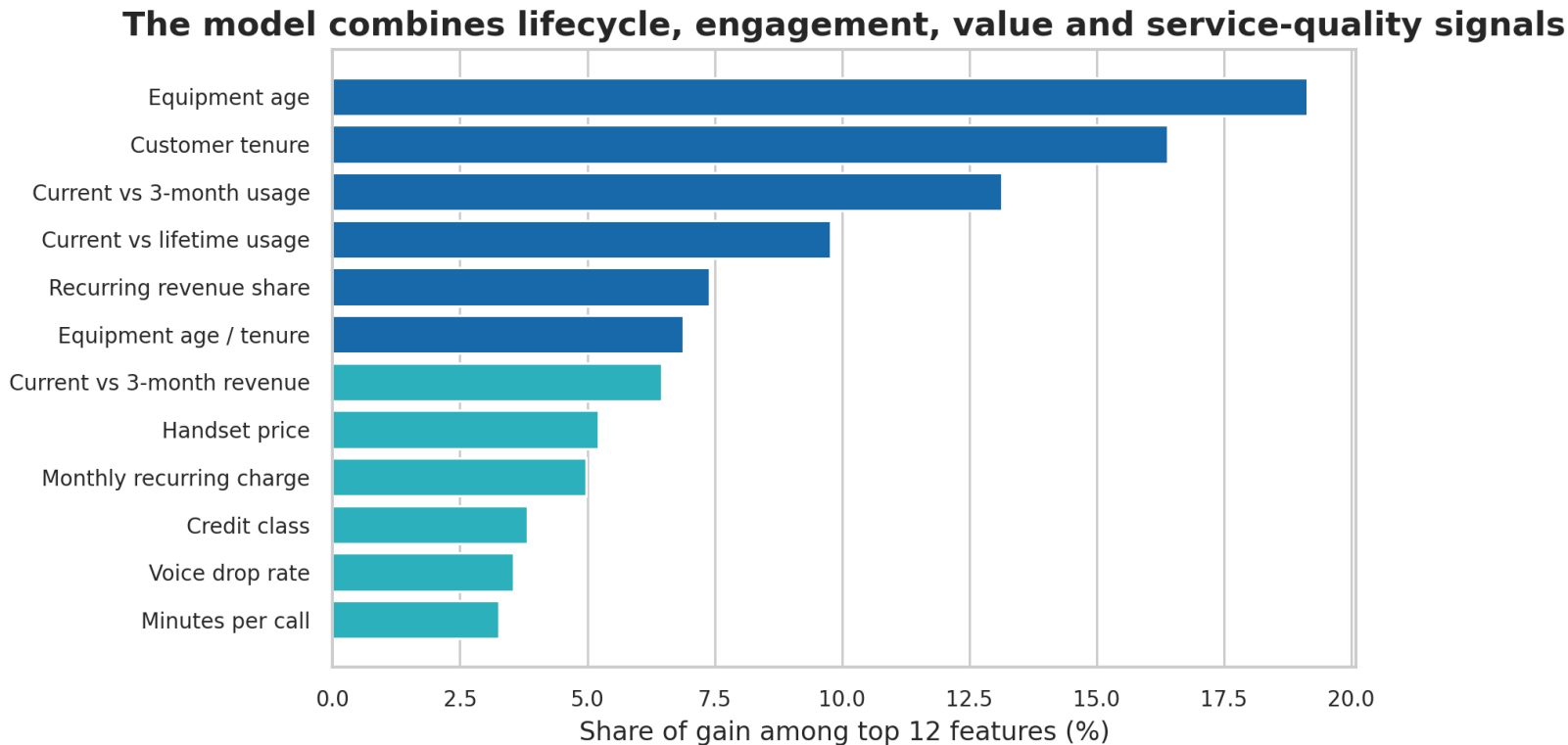
0.702
ROC-AUC

0.695
PR-AUC

64.3%
Accuracy

Use ranking, not accuracy alone

Churn risk is driven by interpretable business signals



Four intervention themes

Lifecycle

Equipment age and tenure

Engagement

Current vs historical usage

Value

Recurring revenue and handset value

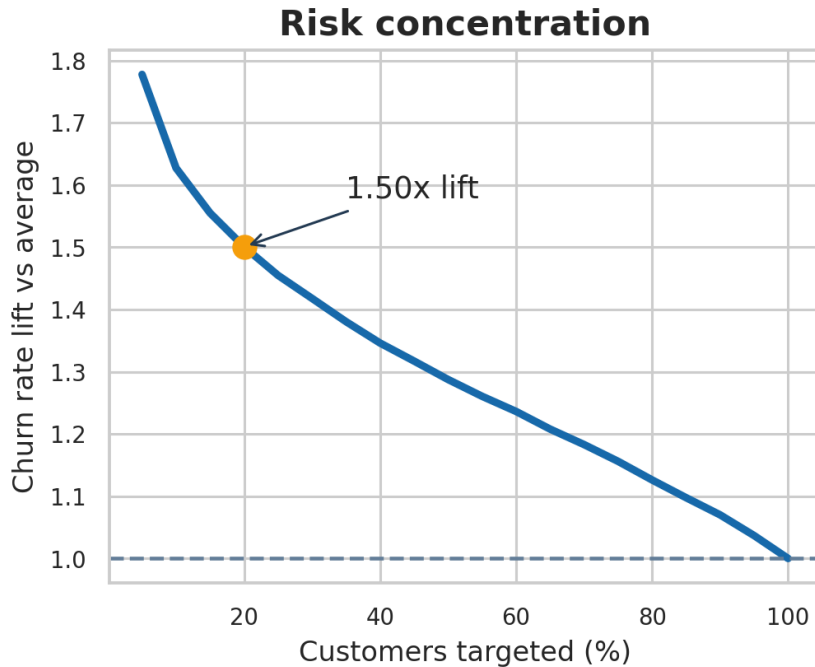
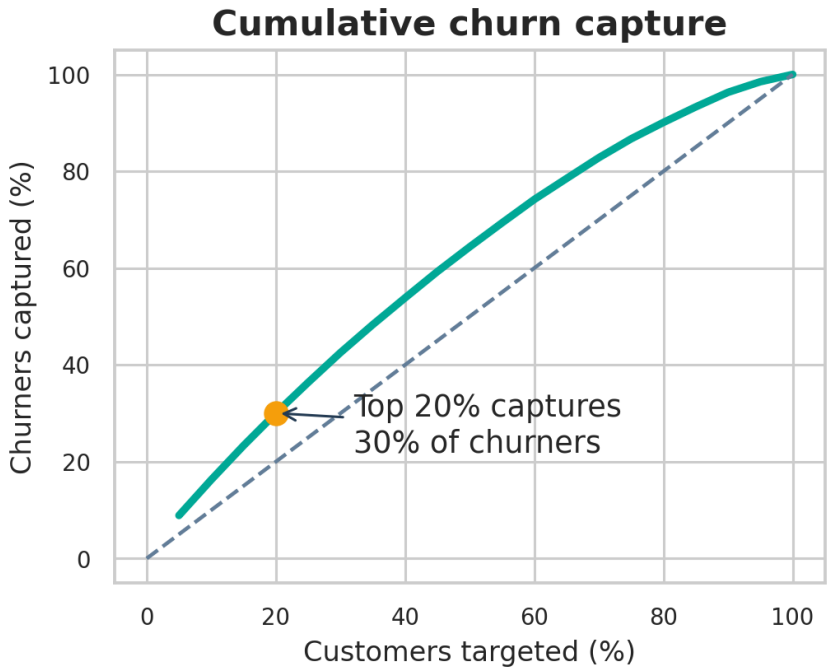
Experience

Voice drops and call completion

Interpretation guardrail

Importance identifies prediction signals, not causal effects.

A top-20% strategy concentrates campaign spend



**Recommended
PoC capacity**

20%

of customers

74.4%

observed churn rate

1.50x

lift vs portfolio

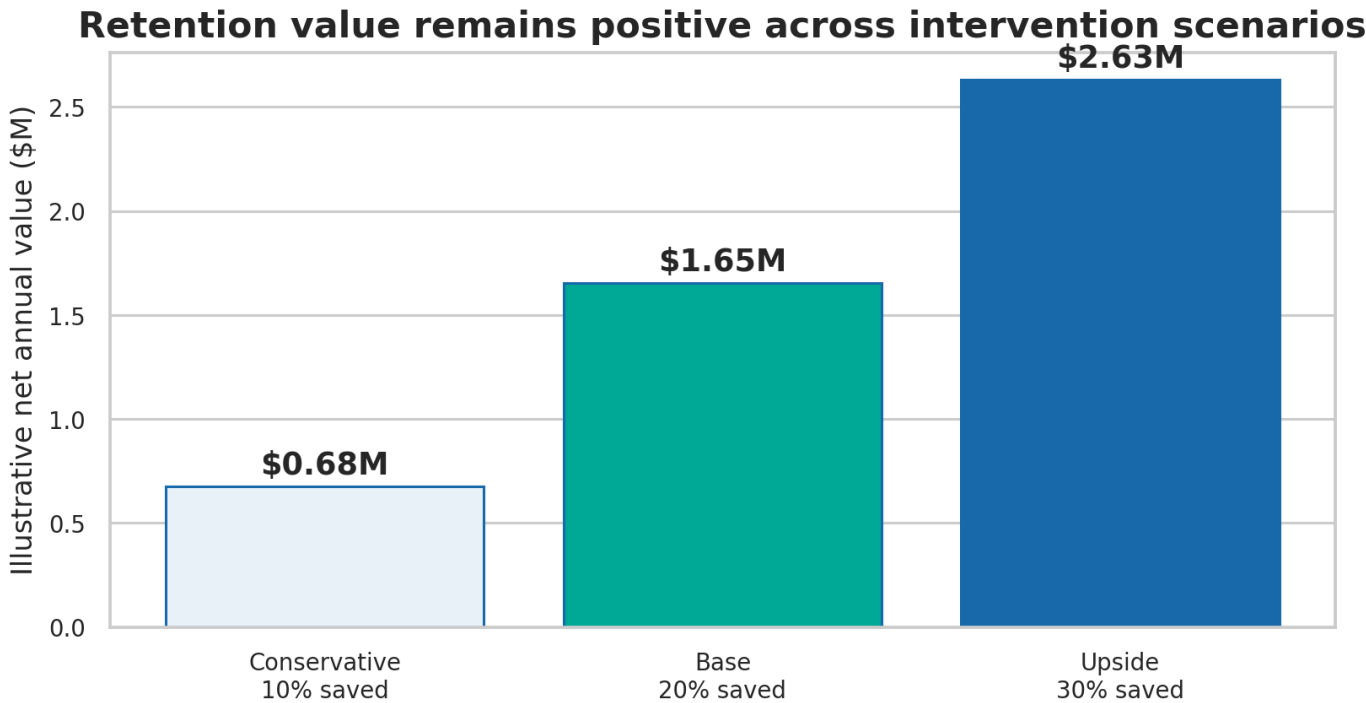
Business proposal: match risk signals to retention actions

Risk pattern	Recommended action	Primary KPI
Ageing handset + high risk	Upgrade offer or device-financing review	Retention uplift
Usage decline	Plan-fit review, personalized bundle or usage benefit	Usage recovery
Service-quality friction	Proactive care call and network/service resolution	Complaint reduction
High-value at-risk customer	Priority outreach with value-capped incentive	Net revenue retained

PoC design: randomly hold out 10% of each risk/action segment to measure incremental retention, not just response rate.

Illustrative economics support a controlled rollout

Scenario per 100,000 scored customers; replace assumptions with Company A campaign costs



Base-case assumptions

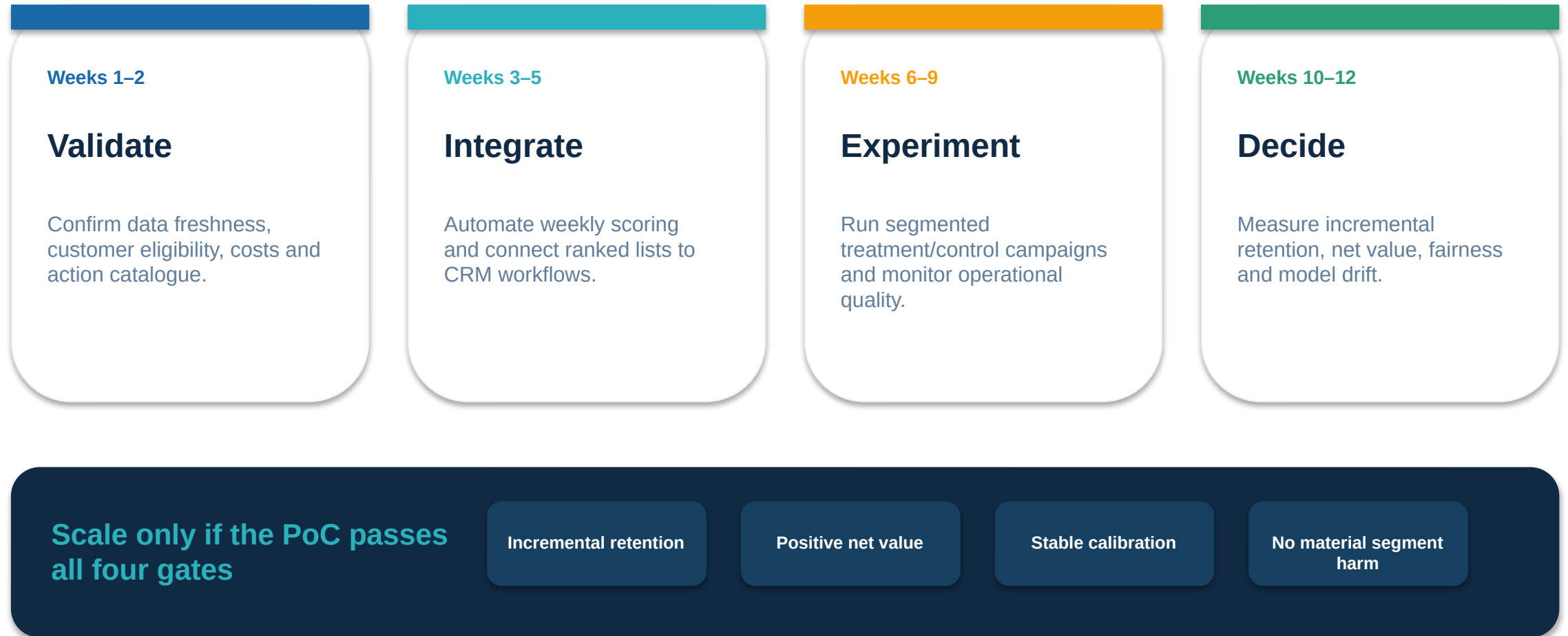
- Target top 20% = 20,000 customers
- Capture 30.0% of expected churners
- Average target revenue = \$54.74/month
- 20% of contacted churners retained
- \$15 intervention cost per targeted customer
- 12 months of retained revenue value

Net annual value

\$1.65M

5.5× ROI

90-day PoC roadmap with decision gates



Decision requested

Approve a 90-day retention PoC targeting the highest-risk 20% of eligible customers, with randomized controls and value-capped interventions.

Expected PoC outputs

- Incremental churn reduction by segment and action
- Verified campaign ROI and scalable capacity
- Production monitoring and retraining requirements

References

[1] Deloitte. 2026 Telecommunications Industry Outlook.
<https://www.deloitte.com/us/en/insights/industry/technology/technology-media-and-telecom-predictions/2026/telecommunications-industry-outlook.html>

[2] Company A Dataset Overview. ENG_Company.md. Supplied course material.

[3] GCI World 2026 Final Assignment Tutorial. tutorial.ipynb. Supplied course material.

[4] Company A analysis pipeline: train_models.py, train_improved_model.py, train_advanced_model.py.

[5] Held-out metrics and figures: artifacts/advanced/metrics.json and feature_importance.csv.

[6] Sample presentation used for narrative reference: Sample1.pptx.

All financial values on slide 13 are explicitly labeled scenario assumptions.